

# Mihan (مِهَن) — Sourced Research Brief v5

Risk-Adjusted Freelancer Credit Engine for Saudi Commercial Banks | AMAD Hackathon 2026

**Primary Track:** المصرفية المفتوحة (Open Banking) **Secondary Tracks:** الذكاء الاصطناعي التوليدي للتقنية المالية • التشريعات المالية • التعليم المالي

Mihan touches four of the six hackathon tracks simultaneously. Register under Open Banking. In the pitch: "هذا المشروع يجمع بين المصرفية المفتوحة والذكاء الاصطناعي والتشريعات المالية والتعليم المالي في منتج واحد."

## 1. Can commercial banks lend to freelancers?

**Answer:** Yes, it is legal. But no commercial bank currently serves actual freelancers.

### 1.1 The critical distinction: "Self-Employed Professionals" vs "Freelancers"

**Self-employed professionals (what Riyadh Bank serves):** Doctors running private clinics, lawyers with their own firms, architects with studios. These individuals have commercial registers, stable high revenue, sometimes employees. They are "self-employed" in the business-owner sense. Financially indistinguishable from salaried employees.

**Actual freelancers (what Mihan serves — unserved 1.6 million):** Graphic designers, content creators, delivery drivers, translators, photographers, marketers, and 110+ other professions registered under the Freelance Work Document. No commercial register, variable income, project-to-project work. No commercial bank has a product or assessment model for this population.

### 1.2 What Riyadh Bank actually offers

| Requirement          | Riyadh Bank                                                                     | Why this excludes actual freelancers      |
|----------------------|---------------------------------------------------------------------------------|-------------------------------------------|
| Eligible professions | 6 only: Architects, Doctors, Lawyers, Engineers, Legal Accountants, Pharmacists | Licensed practice owners, not gig workers |
| Minimum income       | SAR 30,000/month                                                                | 2-3x what most freelancers earn           |
| Service period       | 24+ months                                                                      | Excludes newer freelancers                |
| Process              | Physical branch visit, paper form                                               | No digital application, doesn't scale     |
| Duration             | 36 months max                                                                   | Limited flexibility                       |

| Requirement | Riyad Bank                                               | Why this excludes actual freelancers |
|-------------|----------------------------------------------------------|--------------------------------------|
| Nationality | Saudis only (non-Saudis: Gold/Diamond/Private tier only) | Further narrows pool                 |

**Source:** Riyadh Bank — <https://www.riyadbank.com/personal-banking/personal-finance/self-employed>

**SAB** also offers non-salary-transfer finance (Tawarruq structure). **Source:** SAB — <https://www.sab.com/en/personal/finance/non-salary-transfer-personal-finance/>

### 1.3 What this means for Mihan

1. The legal pathway is proven (Riyad Bank, SAB use Tawarruq structures).
2. Existing products serve a different market (rich professionals, not freelancers).
3. Alinma has no equivalent product — Riyadh Bank is ahead, Mihan leapfrogs.

### 1.4 The pitch line

"Riyad Bank lends to doctors and lawyers making SAR 30,000 a month. That's not freelancing — that's running a business. 1.6 million actual freelancers have zero access. Mihan changes that."

## 2. Regulatory landscape

### 2.1 SAMA's Consumer Financing Regulations

"The Creditor must calculate the Amount of Financing based on the Borrower's **Gross Salary** or monthly pension entitlement."

"Gross Salary" = income paid by an employer monthly.

**Source:** SAMA Rulebook — <https://rulebook.sama.gov.sa/en/regulations-consumer-financing>

**Key insight:** Freelancers have no Gross Salary. SAMA's formula doesn't define an alternative for self-employed individuals. This is a regulatory grey area — but Riyadh Bank and SAB have received SAMA approval for alternative methods (likely bank-statement-based), proving flexibility exists. The specific alternative calculation is not in any public document; it's likely in private SAMA compliance agreements.

### 2.2 SAMA's Open Banking Framework — what it actually is

#### What is Open Banking?

Open Banking is a system where you, the customer, can choose to share your bank data with authorized third parties — securely, through standardized APIs, with your explicit consent, and

under SAMA's supervision. You control what data is shared, with whom, and for how long. You can revoke access at any time.

### **What data can be shared (AIS)?**

Through AIS (Account Information Services), an authorized third party can read:

- Account balances (current and historical)
- Transaction history (deposits, withdrawals, transfers — with dates, amounts, descriptions)
- Recurring payment patterns (subscriptions, bills, rent)
- Income deposits (salary transfers, client payments, freelance income)
- Account metadata (account type, currency, status)

### **Regulatory status of using Open Banking for credit scoring:**

This is the most important regulatory nuance in Mihan's design. Credit scoring is NOT yet explicitly listed as a permitted use case in SAMA's current Open Banking framework (AIS release Nov 2022, PIS release Sep 2024).

AlGhazzawi & Partners (Jun 2025): "Following the rollout of AIS and PIS services, the next phase of Open Banking in Saudi Arabia will **likely include the introduction of broader use cases such as identity verification, credit scoring**, and advanced personal finance management tools."

Riyad Bank lists "Credit Risk Assessment" as an Open Banking service — indicating SAMA intends this but has not yet formally activated it.

**Source:** AlGhazzawi & Partners — <https://www.ghazzawilawfirm.com/insights/open-banking-in-saudi-arabia/> **Source:** Riyadh Bank Open Banking page — <https://www.riyadbank.com/open-banking>

### **The Phase 1 solution — no TPP license needed:**

This is the key insight that makes Mihan immediately viable. Mihan Phase 1 does NOT require Open Banking at all for the core scoring. Here is why:

If the freelancer opens an Alinma business account and routes income through it for 3-6 months, **Alinma already has this transaction data internally**. Alinma does not need Open Banking APIs to access data from its own accounts — it has direct access as the account-holding bank.

Phase 1 uses Alinma's internal transaction data from the freelancer's own Alinma business account. This requires no TPP registration, no Open Banking AIS pull, and no waiting for SAMA to formalize credit scoring as a use case. It is simply a bank analyzing its own customer's transaction history — something banks do today for every account holder.

Phase 2 (cross-bank IBAN matching) is where Open Banking becomes necessary — accessing accounts at other banks requires Mihan to operate as a SAMA-registered TPP. That is a Phase 2 capability, not a day-one requirement.

### **Mihan's two-phase data approach:**

| Phase              | Data source                                                          | Regulatory status                                                        | Requirement                                                   |
|--------------------|----------------------------------------------------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------|
| Phase 1<br>(pilot) | Alinma's own internal data from freelancer's Alinma business account | ✅ No Open Banking needed<br>— Alinma accesses its own data               | Open the Alinma business account, route income for 3-6 months |
| Phase 2<br>(scale) | Cross-bank data via Open Banking AIS (other banks)                   | ⚠️ Requires SAMA TPP registration + credit scoring use case in framework | Post-pilot, pending SAMA Open Banking framework update        |

## Timeline:

| Milestone                                                    | Date                     |
|--------------------------------------------------------------|--------------------------|
| Open Banking policy launched                                 | January 2021             |
| First release — AIS APIs                                     | November 2022            |
| Banks/fintechs readiness required; Open Banking Lab launched | Q1 2023                  |
| Second release — PIS APIs                                    | September 2024           |
| 19 fintech startups in sandbox                               | Mid-2025                 |
| Next phase: identity verification, credit scoring use cases  | Upcoming — not yet dated |

## Sources:

- SAMA via Saudi Press Agency — <https://www.spa.gov.sa/en/N2164738>
- AlGhazzawi & Partners legal analysis — <https://www.ghazzawilawfirm.com/insights/open-banking-in-saudi-arabia/>
- SAMA Official Open Banking page — <https://openbanking.sa/index-en.html>
- Riyadh Bank Open Banking page — <https://www.riyadbank.com/open-banking>

## 2.3 Freelancer commercial bank accounts (SAMA Rule 300-1-1-4)

SAMA Rule No. (300-1-1-4) **mandates** all banks and financial institutions to accept the opening of bank accounts for Freelance Work Document holders — within **one working day** of completing requirements. This is not permissive guidance; it is a mandatory obligation on all Saudi banks.

"SAMA emphasizes that all banks and financial institutions must accept the opening of bank accounts for holders of the Freelance Work Document in accordance with Rule No. (300-1-1-4) of the Rules for Bank Accounts, whether through branches or electronic services."

## **The account's purpose IS the filter.**

A freelancer business account is specifically designed to receive freelance work income. By routing all client payments exclusively through this account, the account structure itself separates freelance income from personal transfers, family money, and unrelated deposits. There is no need for manual tagging to determine what is and isn't income — the account's purpose does that by design.

Everything that flows into the Alinma freelancer business account = verified freelance income. This solves the "messy data" problem cleanly and is more reliable than any algorithmic categorization.

## **What the account doesn't solve — and where Mihan adds intelligence:**

The account tells Mihan HOW MUCH income arrives. It doesn't tell Mihan WHO sent it and whether it's the same client repeatedly or different clients each time. A SAR 8,000 deposit from "Mohammed Al-Harthi" — is that a consistent monthly retainer from one long-term client, or a different person each time?

This is where Mihan applies automatic client pattern detection. Over 3-6 months, Mihan groups recurring deposits by sender identity and payment rhythm. It surfaces to Khalid: "يبدو أن عندك ٣ عملاء" "ثابتين — هل هذا صحيح؟" One confirmation tap. The client portfolio is built passively, not manually.

This feeds directly into the client diversity scoring factor — not just "how much income," but "from how many distinct recurring sources." That's a far more reliable creditworthiness signal than income amount alone.

## **Phase 1 data reality and three paths to solve it:**

Without Open Banking approval, Phase 1 is limited to Alinma's internal data from the new freelancer business account — starting with only 3-6 months of history. This is a genuine limitation on scoring quality. Three paths address it without requiring Open Banking approval:

### **Path 1 — Traditional bank statement submission (available immediately, no regulatory barrier)**

The freelancer voluntarily uploads PDF bank statements from their other banks — Al Rajhi, SNB, wherever they've been banking. Mihan parses those PDFs to extract income deposits and recurring sender IBANs. No Open Banking APIs. No TPP registration. The exact same process Riyadh Bank already requires for their self-employed loan product — standard banking practice globally and in Saudi Arabia.

This gives Mihan income history going back as far as the freelancer's oldest account — potentially 12-24 months — from day one. Limitation: PDFs are easier to manipulate than API data, so confidence is lower. But it is substantively better than zero historical context, and every Saudi bank already accepts this.

**Source:** Riyad Bank Self-Employed Loan (requires manual bank statement submission) — <https://www.riyadbank.com/personal-banking/personal-finance/self-employed>

### **Path 2 — Alinma's existing personal account data (available immediately)**

Many freelancers opening a new Alinma business account already have an Alinma personal account. Alinma holds years of that personal account's transaction history. With the freelancer's explicit consent, Alinma analyzes deposit patterns on the personal account to identify recurring income from the same sender IBANs later confirmed as business clients.

This is NOT Open Banking. Alinma is a single bank analyzing its own customer's data across two accounts it already holds internally. No external APIs, no TPP registration, no regulatory barrier — with one critical requirement.

**PDPL warning — actively enforced:** Using a customer's personal account data for a different product (business loan assessment) is a cross-product data use. The customer opened the personal account under an Account Opening Agreement that did not contemplate credit assessment for a separate product. Under the PDPL's purpose limitation principle, data may only be used for the purpose for which it was collected.

This is not theoretical. SDAIA issued 48 enforcement decisions in 2025 against organizations processing personal data without a valid legal basis. Financial data violations carry fines up to SAR 5 million. Sensitive financial data violations add criminal penalties (up to 2 years imprisonment, SAR 3 million fine).

**The fix is simple:** Add one explicit consent checkbox in the loan application flow: "I consent to Alinma analyzing my personal account deposit history to supplement my business account credit assessment for this loan application." This constitutes valid legal basis under the PDPL. Without it, do not access the personal account.

**Source:** A&O Shearman — PDPL enforcement 48 decisions (Apr 2026) — <https://www.aoshearman.com/en/insights/enforcement-of-the-saudi-personal-data-protection-law>  
**Source:** IAPP — SDAIA active enforcement phase (Feb 2026) — <https://iapp.org/news/a/saudi-arabia-s-data-protection-authority-steps-up-enforcement>

### **Path 3 — SAMA Regulatory Sandbox: the structured path to Open Banking approval**

Rather than waiting for SAMA to update the Open Banking framework independently, Alinma applies to SAMA's Regulatory Sandbox to formally test credit scoring as an Open Banking use case, with Mihan as the technology partner.

The Regulatory Sandbox is open throughout the year for licensed financial institutions and startups. SAMA states a 6-month timeline between application receipt and start of the testing phase, with a testing period of 6-12 months.

**Source:** SAMA Regulatory Sandbox Framework — <https://rulebook.sama.gov.sa/en/entiresection/1368> **Source:** Pontinova Law — SAMA sandbox: 6-month review, 6-12 month testing, always-open model — <https://www.pontinova.law/blog-posts/saudi-arabia-vision-2030-regulatory-sandbox>

Alinma as a licensed financial institution has a structurally stronger application than any startup. SAMA has explicitly signaled credit scoring as the next Open Banking use case. An Alinma-led sandbox application to test freelancer credit scoring would almost certainly be accepted — it is exactly the kind of regulated, institution-backed use case SAMA wants to validate before market-wide rollout.

The sandbox grants real Open Banking AIS access in a controlled, SAMA-supervised environment. Post-sandbox success, formal permanent approval follows as a natural progression.

**Source:** SAMA Open Banking Lab — <https://openbanking.sa/index-en.html> **Source:** Brankas — 19 fintechs in SAMA sandbox by mid-2025, refining credit decisioning protocols — <https://blog.brankas.com/inside-saudi-arabia-open-finance-revolution> **Source:** Ozone API — SAMA licensed Tamawal for finance aggregation via sandbox — <https://ozoneapi.com/the-open-finance-tracker/library/ksa-open-banking-standard/>

**Combined data acquisition timeline:**

| Month       | Data available                                                | Source                         |
|-------------|---------------------------------------------------------------|--------------------------------|
| Day 1       | Existing Alinma personal account history (years)              | Path 2 — internal Alinma data  |
| Day 1       | PDF statements from other banks (months or years)             | Path 1 — voluntary submission  |
| Month 3-6   | 3-6 months Alinma business account (clean, account-by-design) | Core prerequisite              |
| Month 6-7   | Sandbox application approved, Open Banking testing begins     | Path 3 — SAMA sandbox          |
| Month 12-18 | Full cross-bank Open Banking AIS access                       | Path 3 — post-sandbox approval |

**Updated Mihan prerequisite:**

1. Open a dedicated Alinma freelancer business account (free, digital)
2. Voluntarily submit PDF bank statements from any other accounts (extends history immediately)
3. Consent to Alinma analyzing their existing Alinma personal account deposit history
4. Route ALL freelance client payments through the Alinma business account going forward
5. Maintain for 3-6 months before formal assessment begins
6. Agree to a contractual standing deduction order as a loan condition

**Alinma already has freelancer customers** who hold these accounts but receive zero lending products. The pitch: *"You already have these customers. You already have their data — years of it*

in your personal account records. You're just not lending to them. Mihan turns data you already own into loans you're currently not making."

2.4 The Freelance Work Document (وثيقة العمل الحر)

Official document issued by the Ministry of Human Resources and Social Development (HRSD) via the Freelance Work Platform. Covers 120+ approved professions. Free, digital, valid for one year (renewable). Issued through the official portal at [freelance.sa](https://freelance.sa). Requirements: Saudi national, age 20-60, active Absher account.

Sources:

- Official Freelance Platform — <https://freelance.sa>
- National Platform service page — <https://my.gov.sa/ar/services/18485>
- Saudipedia — <https://saudipedia.com/article/12477>

3. Evidence that the solution is needed

3.1 Market scale

| Metric                               | Value                            | Source                             | Note                                                                                                         |
|--------------------------------------|----------------------------------|------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Registered on Freelance Platform     | 2.25 million (Sep 2024)          | Argaam / SPA / Future Work Company | Active registrations at a point in time                                                                      |
| Freelance work documents ever issued | 2.4 million (Jul 2025)           | Bahr platform / HRDF               | Cumulative total — includes expired, renewed, and cancelled documents. NOT the same as active registrations. |
| Currently active freelancers         | 1.674 million (2024)             | Al Watan                           | Currently active subset                                                                                      |
| Economic value of self-employment    | SAR 7.4 billion (2021)           | Saudipedia                         | Likely higher by 2024+                                                                                       |
| Growth trajectory                    | 12,502 (2019) → 2.25M (Sep 2024) | Saudipedia                         | 180x growth in 5 years                                                                                       |

**Why 2.358M and 2.25M don't conflict:** The 2.358M figure is cumulative documents ever issued (total count including expired and renewed). The 2.25M figure is active registrations on the platform at a snapshot (September 2024). Some people's documents expired and weren't renewed, some registered multiple documents. Different metrics, different measurements.



## Sources:

- Argaam, Dec 25, 2024 — <https://www.argaam.com/ar/article/articledetail/id/1778410>
- Al Watan, Jul 2025 — <https://www.alwatan.com.sa/article/1167688>
- Saudipedia — <https://saudipedia.com/article/10580>

## 3.2 Industry leaders confirming the gap

**SIMAH's CBO** (Alaa AlMashhadi) at Money20/20 ME, Sep 2025: explored "how alternative data sources and open banking are reshaping assessments of creditworthiness. This has the potential to unlock financial access for gig workers, informal entrepreneurs, SMEs and others who may lack a conventional credit history but demonstrate financial reliability through other data points."

**Source:** Inside Saudi, Oct 1, 2025 — <https://www.insidesaudi.media/articles/2025-10-01-shaping-the-future-of-finance>

**Lean Technologies CEO** (Hisham Al-Falih): "[Open Banking] will soon enable freelancers, long excluded from traditional banking, to access the capital they need to grow."

**Source:** Gulf Business, Sep 16, 2025 — <https://gulfbusiness.com/money20-20-middle-east-kicks-off-in-riyadh/>

**Money20/20 ME** featured "From Open Finance to Fair Finance" — inDrive.Money case study using ride history, income patterns, and platform engagement instead of legacy credit scores.

**Source:** Money20/20 ME 2025 agenda — <https://money2020middleeast.com/2025-agenda>

## 3.3 The SME lending gap

FSDP 2024 Annual Report (the latest available — published July 13, 2025): SME credit share rose to **9.4%** of total private credit in Q4 2024. Vision 2030 target: **20%**. Gap: 10.6 percentage points.

**Source:** SAMA / FSDP 2024 Annual Report — [https://www.vision2030.gov.sa/media/wpsn44ab/fsdp\\_annual-report-2024\\_-en.pdf](https://www.vision2030.gov.sa/media/wpsn44ab/fsdp_annual-report-2024_-en.pdf)

**Note:** No 2025 annual report has been published yet as of May 2026. The FSDP Q1 2025 quarterly newsletter exists but does not contain updated SME lending percentage data.

MSME credit reached **SAR 420.7 billion** (\$112.18B) by Q2 2025, up 37% YoY.

**Source:** Arab News / SAMA, Oct 2025 — <https://www.arabnews.com/node/2619847/business-economy>

---

## 4. Product design (post-stress-testing)

### 4.1 Core output: Risk-adjusted product recommendations

Mihan outputs a complete, bankable product recommendation — not just a score:

"Khalid qualifies for: Car loan, SAR 60,000, 24 months, APR 7.9%, installments calibrated to worst-month  $\times$  80% capacity (SAR 3,200/month)."

### Critical regulatory requirement — SIMAH is mandatory and runs concurrently:

Mihan does NOT replace the SIMAH credit check. Saudi banking regulations require banks to consult licensed credit information companies as part of any lending decision. The full workflow is:

1. Freelancer applies  $\rightarrow$  Alinma pulls **SIMAH report** (mandatory, no change to current process)
2. Simultaneously  $\rightarrow$  **Mihan engine** analyzes 6-12 months of Alinma business account transaction data
3. SIMAH answers: "Has Khalid repaid debt in the past?" (For most freelancers: thin or empty report — no credit history)
4. Mihan answers: "Does Khalid have the income capacity to repay this loan?" (The gap SIMAH cannot fill)
5. Both outputs feed Alinma's credit committee  $\rightarrow$  lending decision made

SIMAH and Mihan are complementary. Mihan fills exactly the gap SIMAH cannot — income capacity assessment for people with no prior credit history.

**Pitch line:** "SIMAH tells you if Khalid has repaid debt before. Mihan tells you if Khalid can repay debt now. For a freelancer with no credit history, SIMAH is silent. We give Alinma a voice."

**Source:** Credit Information Law — <https://rulebook.sama.gov.sa/en/credit-information-law>

**Source:** SAMA Responsible Lending Principles — <https://rulebook.sama.gov.sa/en/responsible-lending-principles-individual-customers-0>

## 4.2 Graduated tiers

| Tier           | Mihan Score      | Profile                                                                               | Lending approach                                                                        |
|----------------|------------------|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Green (75+)    | Low risk         | 12+ months consistent income, multiple clients, 10%+ savings rate, bills paid on time | Standard products at slightly elevated APR                                              |
| Yellow (55-74) | Medium risk      | 6-12 months history, some income variance, fewer clients                              | Conservative: max SAR 30K, 12 months, higher APR, collateral above SAR 15K              |
| Building (<55) | Not yet eligible | New freelancers, single client, high variance                                         | No lending. Score improvement roadmap + Digital Proof of Income only. Future customers. |

### 4.3 Repayment structure — pre-agreed variable amortization

**Regulatory fact-check result: partially compliant, with important constraints.**

SAMA's Consumer Financing Regulations require two things that directly affect repayment structure:

First, the declining balance method is mandatory: the Term Cost must be allocated proportionally to installments based on the remaining loan balance at the start of each period.

Second, and more importantly: the amortization table must specify the payment date, payment amount, term cost amount, and principal amount for every single payment — fixed at contract signing.

**Source:** SAMA Rulebook, Regulations for Consumer Financing — <https://rulebook.sama.gov.sa/en/regulations-consumer-financing>

**Source:** SAMA Rulebook, Key Conditions for Calculating the Early Repayment Amount — <https://rulebook.sama.gov.sa/en/key-conditions-calculating-early-repayment-amount>

**What this rules out:** Dynamic, real-time flexible installments — where the borrower adjusts payments month-to-month based on actual earnings — are **not compliant** under current SAMA regulations. The amortization table cannot change after signing.

**What this potentially permits:** A pre-agreed variable amortization schedule where all payment amounts are determined upfront, agreed upon, and written into the contract at signing. Peak months and slow months are set before the loan starts — not adjusted dynamically afterward.

**Three repayment options for Mihan (revised):**

**Option A — Flat (fully compliant, available today):** Equal installments every month. Standard, SAMA-approved, no regulatory risk. This is what all Saudi banks currently offer.

**Option B — Pre-agreed seasonal schedule (potentially compliant, needs SAMA approval as new product):** At contract signing, Mihan's income analysis suggests a variable schedule — higher installments in detected peak months, lower in slow months — and both parties agree and lock it into the amortization table. Example: "SAR 4,200 in Oct-Mar, SAR 2,800 in Apr-Sep." All amounts fixed upfront. No post-signing changes permitted. This is mathematically compatible with the declining balance method but would require SAMA's explicit approval as a non-standard product structure. No Saudi bank currently offers this. International precedents exist (graduated payment mortgages in the US, balloon payment structures in the UK) but Saudi regulatory precedent is absent.

**Option C — Customer-proposed schedule (same constraints as Option B):** The customer proposes their own seasonal schedule based on their knowledge of their business. Mihan validates it against income history. Bank and customer agree and sign. Same regulatory status as Option B — potentially permissible if approved by SAMA as a product feature, not currently available.

**Honest pitch position:** Option A is available now and sufficient for the pilot. Options B and C are a product roadmap item — part of the "Phase 2 innovation" narrative — subject to Alinma's

compliance team getting SAMA approval for variable amortization schedules. Presenting them as "available from day one" would be inaccurate. Presenting them as "where we go next, pending SAMA approval" is both honest and forward-looking.

#### 4.4 Risk mitigation framework

**Note on Kafalah:** The Kafalah program requires a commercial register, audited financials for 3 years, a permanent location, and positive SIMAH history. Freelancers with only a Freelance Work Document do NOT qualify.

**Source:** Kafalah requirements — <https://www.saib.com.sa/en/kafalah-program>

Mihan uses four mechanisms instead:

##### 1. Worst-Month × 80% Baseline

The maximum monthly installment is calculated from the borrower's worst month in the past 12 months, multiplied by 0.8 (a 20% safety margin below the worst month). If Khalid's worst month was SAR 6,000, his installment capacity is SAR 4,800 — not his SAR 11,200 average.

This accounts for the possibility that a future bad month could be worse than any historical one. It is MORE conservative than income averaging (which Riyadh Bank likely uses) and provides an extra safety buffer.

**Regulatory note:** SAMA's 33.33% repayment cap is defined against "Gross Salary" (employer-paid). The exact regulatory formula for self-employed income is not publicly defined. Riyadh Bank and SAB have private SAMA approval for their methods. Mihan's worst-month × 80% approach is stricter than averaging, positioning it as easier for SAMA to approve. The specific calculation would be submitted to SAMA through Alinma's compliance team during the pilot — following the same process Riyadh Bank already completed.

##### 2. Pilot as "Cheap Data Acquisition"

Phase 1: 500 borrowers, SAR 10-20K each. Total exposure: SAR 5-10M. Alinma's total assets: SAR 307 billion (Q3 2025). The pilot is 0.003% of total assets.

**Source:** Alinma Bank FY2025 Annual Results — <https://www.alinma.com/en/About-the-Bank/The-Bank/News/2026/1/Net-Profit-for-FY-2025>

Alinma's total assets: **SAR 311,067 million** (SAR 311 billion) as of December 31, 2025. Financing portfolio: SAR 229,747 million. A SAR 10M pilot = **0.003%** of total assets.

This pilot generates proprietary training data (12 months of real default/repayment outcomes) that no competitor has. The scoring model transitions from rule-based to ML-trained. Whoever runs this pilot first owns the data moat — training data cannot be bought, only generated through lending.

##### 3. Contractual standing deduction order

As a loan condition, the freelancer agrees to a standing deduction order on their Alinma business account. When client payments arrive, the bank auto-deducts the installment.

Important distinctions:

- This is a **contractual arrangement** (auto-debit), not the regulated salary deduction mechanism. SAMA's deduction framework is tied to employer salary transfers. A standing order on a business account is a standard banking feature with weaker enforcement.
- The freelancer could theoretically redirect income to another account. However, the 3-6 month prerequisite means clients are already set up to pay into Alinma — redirecting requires changing payment details with every client.
- **This protects against convenience default** (forgetting to pay, choosing to spend elsewhere). **It does NOT protect against insolvency default** (genuinely bankrupt). If the freelancer is broke, no deduction mechanism helps. The worst-month × 80% baseline is the mitigation for insolvency risk — by ensuring the installment is well within capacity even in a worst-case scenario.

**Mihan prerequisite — Alinma business account:** To be assessed by Mihan, the freelancer must:

1. Open a dedicated Alinma freelancer business account (free, digital)
2. Route freelance income through it for 3-6 months
3. Agree to the standing deduction order as a loan condition

This is a product design requirement, not a regulatory mandate. It creates: clean data (business-only transactions), customer lock-in (Alinma ecosystem), and the deduction mechanism.

4. Risk-based pricing

| Tier           | Max amount | Max duration | APR             | Collateral             |
|----------------|------------|--------------|-----------------|------------------------|
| Green (75+)    | SAR 80,000 | 24-36 months | Standard + 1-2% | Not required           |
| Yellow (55-74) | SAR 30,000 | 12 months    | Standard + 3-5% | Required above SAR 15K |

Risk-based pricing is confirmed permissible: banks explicitly state APR varies by credit scoring.

5. Kafalah graduation path (future B2C feature)

When Mihan detects 18-24 months of stable high revenue, it prompts the freelancer:

خالد، أدائك المالي ممتاز خلال آخر سنتين. حان الوقت تسجل سجل تجاري. لو سجلت، بتكون مؤهل لبرنامج كفالة — وهذا يعني تمويل أكبر "ه. أضعاف وبنسبة ربح أقل"

This transitions freelancers from informal economy → formal commercial entity → Kafalah eligibility → larger financing. Alignment with Vision 2030's goal of formalizing the economy.

4.5 Digital Proof of Income (شهادة الدخل الرقمية من الإنماء)

**Branding:** Alinma-branded, Mihan-powered. Alinma issues the document; Mihan provides the engine.

**Critical legal framing:** This must be positioned as an **income verification certificate** — NOT a credit report or creditworthiness assessment. The Credit Information Law prohibits unlicensed entities from providing credit information services. Since Alinma (a licensed bank) is the issuer, it falls outside that definition. The document must not contain a credit score or SIMAH data.

**What it contains:** Verified average monthly income (from Alinma's internal transaction analysis), income consistency metrics, Freelance Work Document status, GOSI contribution status. QR-verifiable PDF.

**What it must NOT contain:** A credit score, creditworthiness rating, or any data resembling a credit report — those are credit information services requiring a SAMA Credit Information license.

Usable beyond Alinma lending: apartment rentals, car dealership financing, telecom postpaid plans, government services.

**Source:** Credit Information Law — <https://rulebook.sama.gov.sa/en/credit-information-law>

## 4.6 Scoring methodology

**The weights challenge:** Finding the right weights is the crux of the entire operation. Here's the honest approach:

### **Phase 1 (Hackathon / Pre-pilot): Rule-based with informed initial weights**

Initial weights are inspired by established credit scoring principles — primarily FICO's structure (the global standard) and adapted for freelancer-specific data:

| Factor                                                                                           | Initial weight | Rationale                                                                                                                                                                                                                                                                            | Data source                                                                                                           |
|--------------------------------------------------------------------------------------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Expense discipline (bill payment consistency, expense-to-income ratio)                           | 35%            | FICO weights payment history at 35% — the strongest predictor of repayment globally. For freelancers, consistent bill payment is the closest equivalent to traditional "payment history."                                                                                            | Open Banking AIS: recurring outflows from business account (rent, STC, electricity, SADAD)                            |
| Income stability (coefficient of variation, trend, worst-month ratio)                            | 30%            | Replaces FICO's "amounts owed" (30%). For freelancers, income stability is the primary capacity indicator. Cross-account historical verification extends this analysis backwards — same sender IBANs appearing in previous accounts extend the income timeline from months to years. | Open Banking AIS: deposits in business account + cross-account IBAN matching in previous personal/other-bank accounts |
| Client diversity (number of distinct recurring senders, single-client dependency, client tenure) | 20%            | Multiple clients = lower concentration risk. Client tenure (verified through cross-account history) adds depth: a client paying for 2 years is a stronger signal than one paying for 2 months.                                                                                       | Open Banking AIS: recurring sender IBAN grouping in business account + historical match in previous accounts          |
| Savings behavior (savings rate, minimum balance, consistency)                                    | 15%            | Savings = buffer against income shocks. Even a small consistent savings rate demonstrates financial discipline.                                                                                                                                                                      | Open Banking AIS: balance history in business account                                                                 |

**A note on cross-account data:** The personal account and other-bank accounts are accessed only for INCOMING transfers from verified business sender IBANs — not for spending patterns or personal transactions. Scope is explicitly limited at the consent stage.

**SAMA AI explainability requirement — mandatory:**

SAMA requires AI-driven credit scoring and loan approvals to be explainable and fair. Banks must inform customers when AI is being used to make financial decisions and give viable reasons when an AI system approves or rejects financial requests.

This means the Mihan scoring product must include:

- A plain-Arabic "Why was I declined?" screen showing the top 2-3 contributing factors
- A "Request Human Review" button for any declined application
- Disclosure at the start of assessment that AI is being used
- Full audit logs of every score calculation, accessible to SAMA inspectors

- A Mihan Score card already shows factor breakdowns (Income Stability, Client Diversity, etc.) — this satisfies the explainability requirement. The additions needed are the recourse mechanism and audit infrastructure.

**Source:** SAMA AI compliance requirements — <https://itbutler.sa/blog/sama-compliance-for-artificial-intelligence-ai/> **Source:** SDAIA AI Ethics Framework —

<https://sdaia.gov.sa/en/SDAIA/about/Documents/ai-principles.pdf> They are educated estimates based on:

- FICO's global payment history weighting (35%) as the anchor
- Tala's validation that alternative data achieves 92% repayment rates with ML-tuned weights
- Upstart's evidence that alternative models approve 44% more borrowers with comparable default rates
- Kaleidofin's PAR90 below 2% using non-traditional data
- Plaid's framework identifying income stability, spending patterns, savings, and DSCR as the four key alternative data signals

**Phase 2 (Post-pilot): ML-calibrated weights**

After the 500-person, 12-month pilot, Mihan has real default/repayment data. A logistic regression or gradient boosting model is trained on actual outcomes to calibrate which factors truly predict default in the Saudi freelancer context. Weights shift from educated guesses to empirically validated coefficients. This is exactly how Tala evolved — rule-based → ML-trained over years.

**International benchmarks:**

| Company     | Model type                                    | Data points                                     | Results                                              | Source                               |
|-------------|-----------------------------------------------|-------------------------------------------------|------------------------------------------------------|--------------------------------------|
| Tala        | ML on 250 alternative data points             | Phone data + behavioral data                    | 92% repayment rate, \$1B+ disbursed to 4M+ customers | Penser analysis                      |
| Upstart     | AI-driven, combines traditional + alternative | Education, employment, cashflow + credit bureau | 44.28% more approvals than traditional models        | SSRN academic paper (Aug 2025)       |
| Kaleidofin  | Alternative scoring for women/underserved     | Financial behavior + non-traditional            | 20-30% higher approval rates, PAR90 below 2%         | SSRN academic paper (Aug 2025)       |
| Spring Bank | Employment-based (no FICO)                    | Employment verification + income                | Total losses under 5% on uncollateralized loans      | Independent Banker / ICBA (Nov 2024) |



## Sources:

- Penser (Tala analysis) — <https://www.penser.co.uk/article/credit-scoring-the-unbanked-alternative-solutions-used-in-global-markets/>
  - SSRN paper on Upstart/Tala/Kaleidofin — [https://papers.ssrn.com/sol3/papers.cfm?abstract\\_id=5386246](https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5386246)
  - ICBA (Spring Bank) — <https://www.independentbanker.org/w/alternative-credit-scoring-models-beyond-fico>
  - Plaid framework — <https://plaid.com/resources/lending/alternative-credit-data/>
- 

## 5. How Mihan compares to the status quo

| Dimension            | Riyad Bank Self-Employed Loan                         | Mihan for Alinma                                                                                                  |
|----------------------|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Target population    | Self-employed professionals (practice owners)         | Actual freelancers (Freelance Work Document holders)                                                              |
| Eligible professions | 6                                                     | All 120+                                                                                                          |
| Minimum income       | SAR 30,000/month                                      | No arbitrary threshold — worst-month × 80% capacity                                                               |
| Risk segmentation    | Binary (approved/rejected)                            | Three tiers (Green/Yellow/Building)                                                                               |
| Assessment method    | Manual paper form + branch visit                      | Automated Open Banking analysis across business account + cross-account historical verification via IBAN matching |
| Income history depth | Self-reported (3-year audited statements for Kafalah) | Up to 24+ months verified automatically via cross-account IBAN matching, no self-reporting needed                 |
| Output               | Approved/rejected                                     | Complete product recommendation with risk-adjusted terms + customer-selected flex repayment                       |
| Risk mitigation      | None specified                                        | Worst-month × 80%, standing deduction, collateral for Yellow, pilot architecture                                  |
| Repayment structure  | Fixed monthly                                         | Customer choice: flat, AI-flex, or custom                                                                         |
| Processing time      | Days to weeks                                         | Under 60 seconds                                                                                                  |

---

## 6. Revenue model

Mihan's current model (SAR 50-100 per assessment + SAR 15-30/month subscription) is too thin. The fundamental problem: it prices Mihan like a data utility rather than like the value it creates. Every loan approved through Mihan's engine is a loan that wouldn't exist otherwise. The revenue model should capture a share of that downstream value — not just charge for the API call.

The redesigned model has five streams, each validated by a real-world precedent.

### **Stream 1: Per-assessment fee + funded loan fee — the baseline B2B revenue**

**What it is:** Two fees charged to Alinma per applicant: (a) a small fee every time Mihan runs an assessment, regardless of outcome; and (b) a larger fee only when a loan is approved AND funded.

**Real-world precedent:** FICO's pricing structure charges \$4.95-\$10 per score pull AND a \$33 funded loan fee per borrower after closing. This dual structure is the industry standard for credit scoring services. FICO's revenue model blends high-margin transaction fees from its Scores business with recurring subscription and SaaS income from its Software segment, producing a diversified, high-retention monetization strategy.

#### **Mihan's pricing:**

- Per-assessment fee: SAR 30-50 per pull (every application, approved or declined)
- Per-funded-loan fee: SAR 125-175 per approved and disbursed loan

**Why this works:** The per-pull fee covers the cost of operating the engine. The per-funded-loan fee is where the real revenue is — and it only triggers when Alinma actually makes money. Clean incentive alignment.

#### **Year 1 projection (conservative):**

- 5,000 assessments × SAR 40 = SAR 200K
- 2,500 funded loans × SAR 150 = SAR 375K
- **Stream 1 total: SAR 575K/year**

### **Stream 2: Platform SaaS license — the predictable baseline**

**What it is:** An annual platform license fee covering: unlimited assessments (or a cap), model updates, bank-side dashboard, integration maintenance, compliance documentation, and API access.

**Real-world precedent:** FICO's Software segment — now cloud-native SaaS — generates 44% of total company revenue, with Software ARR growing 20% year-over-year. Zest AI licenses its AI underwriting models to financial institutions as a SaaS platform — Zest AI has a 100% retention

rate for its credit union customers and has built close to 400 AI-underwriting models for financial institutions. Financial institutions implementing Zest AI underwriting solutions can expect to automate 60-80% of lending decisions while reducing charge-offs by 20%.

**Mihan's pricing:** SAR 300K-600K/year annual platform license to Alinma, covering the full engine with a per-assessment cap. Predictable baseline revenue regardless of origination volume.

**Year 1 projection:** SAR 400K/year (negotiated as part of the pilot agreement)

### **Stream 3: Digital Proof of Income — the standalone B2C product**

**What it is:** The شهادة الدخل الرقمية من الإنماء (Alinma-branded, Mihan-powered) is sold as a paid document to freelancers who need to prove income for apartment leases, car dealership financing, telecom postpaid plans, and government services — entirely independent of whether they take a loan.

**Real-world precedent:** Equifax's Work Number charges approximately \$105 per verification as of January 2026 — per applicant, per report. Over 65,000 credentialed verifier organizations trust the service. The Community Home Lenders of America cited the cost at \$55-70 per verification pull, arguing it was "excessive" — which actually demonstrates the pricing power of verified income data.

Mihan's version is cheaper (freelancers self-serve) and differentiated (Open Banking-verified vs. employer-reported), but the market precedent shows income verification commands serious fees.

**Mihan's pricing:** SAR 75 per certificate, valid 90 days. No subscription required — one-time purchase for a specific deliverable.

**Year 1 projection:** 10,000 certificates × SAR 75 = **SAR 750K/year** **Year 3 projection (1% of 1M genuine freelancers):** 100,000 × SAR 75 = **SAR 7.5M/year**

### **Stream 4: Invoicing and payment rails — the long game**

**What it is:** Since Mihan's prerequisite routes freelancer income through Alinma business accounts, a lightweight invoicing tool embedded in the Mihan/Alinma app is a natural next step. Freelancers create invoices, send to clients, receive payment through Alinma. Every paid invoice strengthens the freelancer's Mihan Score automatically. Mihan charges a transaction fee on invoiced payments.

**Real-world precedent:** Stripe charges 2.9% + \$0.30 per transaction. Square charges 2.6% + \$0.10 per swipe. In Saudi Arabia, Hala (a fintech targeting freelancers specifically) offers business accounts and payment processing on a similar fee model. PayTabs Saudi charges 2.5-2.9% per transaction.

**Mihan's pricing:** 0.75% of invoice value processed through Alinma. This is below Stripe/Square (incentivizing use) while still being meaningful at scale.

**Long-term projection (not Year 1):** 100K freelancers × SAR 100K average annual invoicing × 0.75% = **SAR 75M/year** at scale. This is a 3–5 year stream, not a hackathon-day-one feature. Show judges the roadmap.

**Revenue model summary**

| Stream                           | Year 1    | Year 3    | Real-world precedent                          |
|----------------------------------|-----------|-----------|-----------------------------------------------|
| Per-assessment + funded loan fee | SAR 575K  | SAR 3M    | FICO (\$4.95–33 per score/funded loan)        |
| Platform SaaS license (Alinma)   | SAR 400K  | SAR 500K  | Zest AI (SaaS, 100% retention), FICO Software |
| Digital Proof of Income          | SAR 750K  | SAR 7.5M  | Equifax Work Number (\$66–105/verification)   |
| Payment rails (invoicing)        | —         | SAR 15M+  | Stripe (2.9%), Hala Saudi, PayTabs            |
| Total                            | ~SAR 1.7M | ~SAR 26M+ |                                               |

**The pitch framing**

Stop presenting Mihan as a data vendor. Present it as infrastructure:

*"FICO makes \$5 every time a bank checks a credit score — and \$33 every time a loan closes. Equifax charges \$105 every time a landlord verifies someone's income. These are not large fees. But they happen millions of times. Mihan applies the same model to the 1.6 million Saudi freelancers who've never had a credit score. We charge SAR 40 per assessment and SAR 150 per funded loan. We sell income certificates at SAR 75. After 18 months, we license to every Saudi bank. That's a platform, not a feature."*

**7. Market sizing (conservative)**

**TAM — Total Addressable Market**

1.674M active freelancers. Conservative 60% genuine active rate = ~1M.

| Revenue stream                      | Calculation                                      | Value                 |
|-------------------------------------|--------------------------------------------------|-----------------------|
| B2B assessment fees                 | $1\text{M} \times \text{SAR } 75$                | ~SAR 75M/year         |
| B2C subscriptions (15%)             | $150\text{K} \times \text{SAR } 200/\text{year}$ | ~SAR 30M/year         |
| B2C Proof of Income (30%)           | $300\text{K} \times \text{SAR } 50/\text{year}$  | ~SAR 15M/year         |
| <b>Total TAM</b>                    |                                                  | <b>~SAR 120M/year</b> |
| Downstream lending (25% qualifying) | $250\text{K} \times \text{SAR } 50\text{K}$      | <b>~SAR 12.5B</b>     |

### SAM — Serviceable Addressable Market

~400K freelancers with SAR 4K+/month income and 6+ months history.

| Revenue stream                  | Value                  |
|---------------------------------|------------------------|
| <b>Total SAM</b>                | <b>~SAR 47.5M/year</b> |
| Downstream at Alinma's 8% share | <b>~SAR 1.6B</b>       |

### SOM — Year 1 (pilot + initial rollout)

| Revenue stream                                        | Value                 |
|-------------------------------------------------------|-----------------------|
| <b>Total SOM</b>                                      | <b>~SAR 775K/year</b> |
| Pilot lending ( $500 \times \text{SAR } 40\text{K}$ ) | <b>SAR 20M</b>        |
| Post-pilot rollout                                    | <b>SAR 225M</b>       |

## 8. Risk mitigation summary

| Risk                   | Mitigation                                                                                                                               | Severity |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Higher default rates   | Worst-month $\times$ 80% baseline; risk-based pricing; graduated tiers (only Green/Yellow lend)                                          | Managed  |
| No training data       | Rule-based Phase 1 $\rightarrow$ 500-person pilot (SAR 5-10M, 0.003% of Alinma's SAR 311B assets) generates proprietary ML training data | Managed  |
| Messy transaction data | Dedicated Alinma business account (income-pure by design); composite 4-factor scoring                                                    | Managed  |

| Risk                                             | Mitigation                                                                                                                                                                                                                    | Severity |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Kafalah unavailable for freelancers              | Risk mitigated through worst-month $\times$ 80%, standing deduction, collateral. Kafalah repositioned as 3-year graduation path.                                                                                              | Managed  |
| Convenience default                              | Contractual standing deduction on business account (CONFIRMED LEGAL — prior written consent at signing = valid under SAMA rules)                                                                                              | Managed  |
| Insolvency default                               | Worst-month $\times$ 80% ensures installments are within capacity even in worst scenarios. Collateral for Yellow above SAR 15K.                                                                                               | Managed  |
| 33.33% cap — no public formula for self-employed | <b>CRITICAL PILOT BLOCKER. Alinma compliance must submit Mihan's income calculation methodology to SAMA before issuing any live loans.</b> Worst-month $\times$ 80% is stricter than averaging — should be easier to approve. | Critical |
| PDPL data residency — criminal liability         | <b>All processing must occur on Saudi-hosted infrastructure. Violation: up to 1 year imprisonment + SAR 1M fine.</b> Must be verified before any real user data is processed.                                                 | Critical |
| PDPL cross-product consent                       | Personal account data (Path 2) requires explicit consent checkbox. SDAIA issued 48 enforcement decisions in 2025 — actively enforced. Violation: up to SAR 5M fine.                                                           | High     |
| PDPL DPIA + DPO                                  | Required before deployment of automated financial decision-making at scale. Criminal penalties for sensitive data violations.                                                                                                 | High     |
| AI explainability (SAMA requirement)             | Factor breakdown already satisfies explainability. Add: decline reasons screen in Arabic, human review button, audit logs.                                                                                                    | Medium   |
| Open Banking credit scoring not yet formal       | Phase 1 uses internal data only (no Open Banking needed). Sandbox application is the path to Phase 2.                                                                                                                         | Managed  |
| SIMAH / Lean competition                         | SIMAH = data infrastructure. Lean = API rails. First-mover pilot data is the moat.                                                                                                                                            | Low      |

## 9. Anticipated judge objections

**"Irregular income IS higher risk."** Installments are anchored to worst-month  $\times$  80% — not the average. The repayment is calibrated to a scenario worse than the borrower's actual worst month. Risk-based pricing (higher APR, shorter terms) compensates for elevated probability. A contractual standing deduction captures installments from incoming deposits, protecting against convenience default.

**"SDB already lends to freelancers."** SDB provides startup capital — up to SAR 200K for launching a business. That's not a car loan, credit card, or personal finance for daily life. Every consumer banking product a freelancer needs remains inaccessible from commercial banks. SDB and Mihan serve different needs entirely.

**"Won't SIMAH build this?"** SIMAH's own CBO acknowledged this gap at Money20/20 in September 2025. It's May 2026 — nothing shipped. SIMAH is data infrastructure, not product innovation. A generic SIMAH model would be available to all banks with no competitive advantage for Alinma. Mihan gives Alinma a proprietary first-mover position.

**"How do you know the model works with no data?"** Phase 1 is rule-based, with weights informed by FICO's global structure and validated by companies like Tala (92% repayment rate) and Upstart (44% more approvals). The 500-person pilot is how we validate for the Saudi context. Whoever runs this pilot first owns the data moat — training data cannot be bought, only generated by lending.

**"What about defaults?"** Three layers: (1) graduated tiers — only Green and Yellow freelancers get loans; (2) worst-month  $\times$  80% baseline — installments calculated against a scenario worse than the borrower's actual worst month; (3) contractual standing deduction on the Alinma business account captures installments from incoming deposits. The pilot is SAR 10M — 0.003% of Alinma's SAR 311B assets.

**"What about the standing deduction if someone goes bankrupt?"** The standing deduction mitigates convenience default — forgetting to pay, choosing to spend elsewhere. It does NOT protect against genuine insolvency. Insolvency risk is mitigated by the worst-month  $\times$  80% baseline, which ensures the installment is dramatically below capacity even in a worst case. The pilot's small scale means even worst-case losses are contained at a rounding-error level for Alinma.

**"Is the flexible repayment plan actually legal?"** Option A (flat equal installments) is fully SAMA-compliant and what we launch with. Options B and C (pre-agreed seasonal schedules) are a Phase 2 product roadmap item — mathematically compatible with the declining balance method but requiring SAMA approval as a non-standard amortization structure. We present them as innovation pipeline, not day-one features.

## 10. Critical pre-pilot requirements

Four items must be completed before any real user data enters the system. These do not block the hackathon demo (which uses synthetic data), but they are non-negotiable before the 500-person pilot launches.

### 10.1 — SAMA approval for alternative income calculation methodology (CRITICAL PILOT BLOCKER)

This is the single most important pre-pilot action. SAMA's Consumer Financing Regulations calculate repayment capacity from "Gross Salary" — defined as employer-paid income. Freelancers have no Gross Salary. The 33.33% deduction cap therefore has no defined calculation method for freelancers in any public document.

Before Alinma issues a single live loan through Mihan, Alinma's compliance team must submit Mihan's Worst-Month  $\times$  80% income calculation methodology to SAMA and receive formal approval. Riyadh Bank and SAB have already received private SAMA approvals for their own alternative methods — the precedent exists. But Alinma cannot assume their methodology is automatically approved by analogy. This is a regulatory process with an undefined timeline that must begin immediately after the hackathon.

**Source:** SAMA Consumer Financing Regulations — <https://rulebook.sama.gov.sa/en/regulations-consumer-financing>

## **10.2 — Saudi-hosted cloud infrastructure (CRIMINAL LIABILITY)**

Saudi Arabia's PDPL and SDAIA's cross-border transfer regulation (September 2024) require that sensitive personal data be processed within the Kingdom. Financial transaction data is sensitive personal data. Unauthorized transfer of sensitive data outside Saudi Arabia carries: up to SAR 1 million fine plus up to one year imprisonment.

All Mihan data processing must occur on Saudi-region cloud infrastructure (AWS Bahrain/Riyadh, Azure Saudi North, or equivalent). This is a technical architecture decision that must be confirmed before any real user data enters the system.

**Source:** Global Privacy Blog — <https://www.globalprivacyblog.com/2024/03/six-months-until-enforcement-key-compliance-steps-for-saudi-arabias-data-protection-law/> **Source:** SDAIA Data Transfer Regulation (September 2024)

## **10.3 — DPIA and DPO appointment**

The PDPL requires a Data Protection Impact Assessment before deploying any system that processes sensitive personal data at scale for automated decision-making. A Data Protection Officer must be appointed. SDAIA issued 48 enforcement decisions in 2025 — the enforcement phase is active.

**Source:** A&O Shearman — PDPL enforcement 48 decisions — <https://www.aoshearman.com/en/insights/enforcement-of-the-saudi-personal-data-protection-law>

## **10.4 — Cross-product consent for personal account data (Path 2)**

If using the freelancer's Alinma personal account history for loan assessment, a specific consent checkbox must be added to the loan application: "I consent to Alinma analyzing my personal account deposit history to supplement my business account credit assessment for this application." Without this, using personal account data violates the PDPL purpose limitation principle — now actively enforced.

---

## **11. Final verdict — can we build this for Alinma now?**

**The one-line answer**

**Yes — Phase 1 is eligible to build and deploy for Alinma today, under Alinma's existing**



**banking license, without any new regulatory approvals.**

### What is eligible now (Phase 1)

| Feature                                                         | Eligible now?        | Why                                                                                                                                      |
|-----------------------------------------------------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Scoring engine analyzing Alinma's own internal data             | ✓ Yes                | Alinma already has this data. No Open Banking, no TPP license. Internal analytics.                                                       |
| SIMAH check (concurrent, mandatory)                             | ✓ Yes                | Standard process Alinma already runs. Unchanged.                                                                                         |
| Mihan Score + risk-adjusted product recommendation              | ✓ Yes                | Analytical tool for Alinma's internal use. Bank outsourcing framework permits this.                                                      |
| Flat installment repayment (Option A)                           | ✓ Yes                | Standard SAMA-compliant structure. No changes needed.                                                                                    |
| Risk-based APR by tier                                          | ✓ Yes                | Explicitly confirmed permissible. Banks vary APR by credit profile.                                                                      |
| Freelancer business account as prerequisite                     | ✓ Yes                | Enabled by SAMA Rule 300-1-1-4. Alinma already supports it.                                                                              |
| Contractual standing deduction order                            | ✓ Yes                | Standard auto-debit contract term. Not the regulated salary deduction — weaker, but permissible.                                         |
| Worst-Month $\times$ 80% as internal risk methodology           | ✓ Yes (with caveat)  | Internal methodology. The official income formula for self-employed must be validated with Alinma's compliance team before pilot launch. |
| 500-person pilot under Alinma's license                         | ✓ Yes                | No special SAMA approval needed for a bank running an internal product pilot.                                                            |
| Digital Proof of Income (income certificate, not credit report) | ✓ Yes (with framing) | Issued by Alinma as income verification. Must NOT contain credit score or SIMAH data.                                                    |
| B2C premium features (score improvement, income forecasting)    | ✓ Yes                | Customer-facing analytics within Alinma's app. No additional license.                                                                    |
| Kafalah graduation path notification                            | ✓ Yes                | Informational feature. No financial service being provided.                                                                              |

## What is NOT eligible now (Phase 2)

| Feature                                      | Why not yet                                                                                       | What needs to happen                                  |
|----------------------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| Cross-bank IBAN matching (other banks' data) | Requires SAMA TPP registration + credit scoring use case formally added to Open Banking framework | SAMA TPP registration + await next Open Banking phase |
| Open Banking AIS credit scoring (explicit)   | Credit scoring not yet formally listed in current framework                                       | SAMA Open Banking framework update (anticipated)      |
| Pre-agreed seasonal repayment (Options B/C)  | No Saudi bank currently offers variable amortization                                              | Alinma compliance → SAMA product approval             |

## The honest pitch to Alinma judges

*"Everything in Phase 1 is within Alinma's existing regulatory authority. We analyze your own customers' transaction data from their Alinma business accounts — data you already have. We run SIMAH alongside us as required. We operate as your outsourced analytics vendor under standard SAMA outsourcing rules. We pilot with 500 freelancers under your existing banking license. No new SAMA approval needed to start. The pilot itself generates the data that makes Phase 2 stronger."*

*"Phase 2 — cross-bank data, full Open Banking credit scoring, variable amortization — those require SAMA approvals we've identified specifically. All three have clear regulatory pathways. We're not guessing at regulation. We know exactly what needs to happen and in what order."*

## Why this matters for the hackathon

Most teams pitching at AMAD 2026 will not have done this level of regulatory analysis. They will say "we use Open Banking" without knowing that credit scoring via Open Banking is not yet explicitly permitted. They will say "Kafalah reduces our risk" without knowing freelancers don't qualify. They will propose flexible installments without knowing SAMA requires fixed amortization tables.

Mihan's competitive advantage in the pitch room is not just the product — it's the team that did the regulatory homework before building. That signals to Alinma judges that this team understands the environment they'd be deploying into.

**Source:** Full regulatory verification in companion document:  
Mihan\_Regulatory\_Verification.md

## 12. Source index — all URLs verified May 17, 2026

| # | What it sources                                                                             | Publisher                     | URL                                                                                                                                                                                                                             |
|---|---------------------------------------------------------------------------------------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | SAMA<br>Consumer<br>Financing<br>Regulations<br>(Gross Salary<br>definition,<br>33.33% cap) | SAMA<br>Rulebook              | <a href="https://rulebook.sama.gov.sa/en/regulations-consumer-financing">https://rulebook.sama.gov.sa/en/regulations-consumer-financing</a>                                                                                     |
| 2 | Riyad Bank Self-<br>Employed Loan<br>(6 professions,<br>SAR 30K min,<br>36mo max)           | Riyad Bank                    | <a href="https://www.riyadbank.com/personal-banking/personal-finance/self-employed">https://www.riyadbank.com/personal-banking/personal-finance/self-employed</a>                                                               |
| 3 | SAB Non-salary<br>Transfer<br>Finance<br>(Tawarruq<br>structure)                            | SAB                           | <a href="https://www.sab.com/en/personal/finance/non-salary-transfer-personal-finance/">https://www.sab.com/en/personal/finance/non-salary-transfer-personal-finance/</a>                                                       |
| 4 | Open Banking<br>PIS release<br>announcement                                                 | Saudi Press<br>Agency         | <a href="https://www.spa.gov.sa/en/N2164738">https://www.spa.gov.sa/en/N2164738</a>                                                                                                                                             |
| 5 | Open Banking<br>legal analysis for<br>Saudi Arabia                                          | AlGhazzawi<br>& Partners      | <a href="https://www.ghazzawilawfirm.com/insights/open-banking-in-saudi-arabia/">https://www.ghazzawilawfirm.com/insights/open-banking-in-saudi-arabia/</a>                                                                     |
| 6 | Open Banking<br>PIS framework<br>details                                                    | Open<br>Banking<br>Expo       | <a href="https://www.openbankingexpo.com/news/saudi-central-bank-issues-second-release-of-open-banking-framework/">https://www.openbankingexpo.com/news/saudi-central-bank-issues-second-release-of-open-banking-framework/</a> |
| 7 | SAMA Official<br>Open Banking<br>Program page                                               | SAMA                          | <a href="https://openbanking.sa/index-en.html">https://openbanking.sa/index-en.html</a>                                                                                                                                         |
| 8 | Freelance Work<br>Document —<br>official portal                                             | HRSD /<br>freelance.sa        | <a href="https://freelance.sa">https://freelance.sa</a>                                                                                                                                                                         |
| 9 | Freelance Work<br>Document —<br>National                                                    | Saudi<br>National<br>Platform | <a href="https://my.gov.sa/ar/services/18485">https://my.gov.sa/ar/services/18485</a>                                                                                                                                           |

| #  | What it sources                                        | Publisher                         | URL                                                                                                                                                                           |
|----|--------------------------------------------------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | Platform service page                                  |                                   |                                                                                                                                                                               |
| 10 | Freelance Work Document — program details              | Saudipedia                        | <a href="https://saudipedia.com/article/12477">https://saudipedia.com/article/12477</a>                                                                                       |
| 11 | 2.25M platform registrations (Sep 2024)                | Argaam / Saudi Press Agency       | <a href="https://www.argaam.com/ar/article/articledetail/id/1778410">https://www.argaam.com/ar/article/articledetail/id/1778410</a>                                           |
| 12 | 1.674M active freelancers (2024)                       | Al Watan                          | <a href="https://www.alwatan.com.sa/article/1167688">https://www.alwatan.com.sa/article/1167688</a>                                                                           |
| 13 | 2.4M freelance documents total (Jul 2025)              | Bahr Platform / HRDF via Al Watan | <a href="https://www.alwatan.com.sa/article/1167171">https://www.alwatan.com.sa/article/1167171</a>                                                                           |
| 14 | SAR 7.4B economic value of self-employment             | Saudipedia                        | <a href="https://saudipedia.com/article/10580">https://saudipedia.com/article/10580</a>                                                                                       |
| 15 | SIMAH CBO at Money20/20 ME on gig worker credit gap    | Inside Saudi                      | <a href="https://www.insidesaudi.media/articles/2025-10-01-shaping-the-future-of-finance">https://www.insidesaudi.media/articles/2025-10-01-shaping-the-future-of-finance</a> |
| 16 | Lean Technologies CEO on freelancer banking access     | Gulf Business                     | <a href="https://gulfbusiness.com/money20-20-middle-east-kicks-off-in-riyadh/">https://gulfbusiness.com/money20-20-middle-east-kicks-off-in-riyadh/</a>                       |
| 17 | Money20/20 ME 2025 "From Open Finance to Fair Finance" | Money20/20 ME                     | <a href="https://money2020middleeast.com/2025-agenda">https://money2020middleeast.com/2025-agenda</a>                                                                         |
| 18 | FSDP 2024 Annual Report — 9.4% SME lending share       | SAMA / Vision 2030                | <a href="https://www.vision2030.gov.sa/media/wpsn44ab/fsdp_annual-report-2024_-en.pdf">https://www.vision2030.gov.sa/media/wpsn44ab/fsdp_annual-report-2024_-en.pdf</a>       |

| #  | What it sources                                                                | Publisher                 | URL                                                                                                                                                                                                                                 |
|----|--------------------------------------------------------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 19 | MSME credit SAR 420.7B (Q2 2025, +37% YoY)                                     | Arab News / SAMA          | <a href="https://www.arabnews.com/node/2619847/business-economy">https://www.arabnews.com/node/2619847/business-economy</a>                                                                                                         |
| 20 | Kafalah program eligibility (commercial register, audited financials required) | Saudi Investment Bank     | <a href="https://www.saib.com.sa/en/kafalah-program">https://www.saib.com.sa/en/kafalah-program</a>                                                                                                                                 |
| 21 | Alinma FY2025 results (SAR 311B assets, SAR 6.4B net profit)                   | Alinma Bank               | <a href="https://www.alinma.com/en/About-the-Bank/The-Bank/News/2026/1/Net-Profit-for-FY-2025">https://www.alinma.com/en/About-the-Bank/The-Bank/News/2026/1/Net-Profit-for-FY-2025</a>                                             |
| 22 | Alinma Q3 2025 interim financial statements (PDF)                              | Alinma Bank IR            | <a href="https://ir.alinma.com/media/3erjpgdv/alinma-fs-en-q3-2025.pdf">https://ir.alinma.com/media/3erjpgdv/alinma-fs-en-q3-2025.pdf</a>                                                                                           |
| 23 | Tala alternative credit scoring (250 data points, 92% repayment rate)          | Penser                    | <a href="https://www.penser.co.uk/article/credit-scoring-the-unbanked-alternative-solutions-used-in-global-markets/">https://www.penser.co.uk/article/credit-scoring-the-unbanked-alternative-solutions-used-in-global-markets/</a> |
| 24 | Upstart (+44% approvals), Tala, Kaleidofin (PAR90 <2%) comparison              | SSRN Academic Paper       | <a href="https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5386246">https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5386246</a>                                                                                               |
| 25 | Spring Bank employment-based lending (losses under 5%)                         | ICBA / Independent Banker | <a href="https://www.independentbanker.org/w/alternative-credit-scoring-models-beyond-fico">https://www.independentbanker.org/w/alternative-credit-scoring-models-beyond-fico</a>                                                   |
| 26 | Plaid framework: income stability, spending, savings, DSCR as key signals      | Plaid                     | <a href="https://plaid.com/resources/lending/alternative-credit-data/">https://plaid.com/resources/lending/alternative-credit-data/</a>                                                                                             |

| #  | What it sources                                                                                        | Publisher                        | URL                                                                                                                                                                                                                                                                                                 |
|----|--------------------------------------------------------------------------------------------------------|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 27 | FICO pricing: \$4.95 per score + \$33 funded loan fee; SaaS ARR +20% YoY                               | National Mortgage Professional   | <a href="https://nationalmortgageprofessional.com/news/ficos-direct-license-program-shakes-mortgage-credit-scoring">https://nationalmortgageprofessional.com/news/ficos-direct-license-program-shakes-mortgage-credit-scoring</a>                                                                   |
| 28 | FICO revenue model: Scores 54% + Software/SaaS 44% of 2024 revenue                                     | Margin of Sanity / FICO analysis | <a href="https://marginofsanity.substack.com/p/fico-what-the-hell-is-going-on-full">https://marginofsanity.substack.com/p/fico-what-the-hell-is-going-on-full</a>                                                                                                                                   |
| 29 | Equifax Work Number: \$66-105 per income verification; 65,000+ verifier organizations                  | Housing Wire / CHLA              | <a href="https://www.housingwire.com/articles/equifaxs-work-number-is-too-expensive-and-regulators-should-look-into-it-chla/">https://www.housingwire.com/articles/equifaxs-work-number-is-too-expensive-and-regulators-should-look-into-it-chla/</a>                                               |
| 30 | Zest AI: 400+ AI underwriting models; 100% retention; 20% approval increase + 20% charge-off reduction | Zest AI press release            | <a href="https://www.prnewswire.com/news-releases/zest-ais-credit-models-proven-to-increase-loan-approvals-for-every-protected-class-302048828.html">https://www.prnewswire.com/news-releases/zest-ais-credit-models-proven-to-increase-loan-approvals-for-every-protected-class-302048828.html</a> |
| 31 | Upstart: \$33B in originations from 99 bank partners; 44% more approvals                               | Credible review                  | <a href="https://www.credible.com/personal-loan/upstart-personal-loans-review">https://www.credible.com/personal-loan/upstart-personal-loans-review</a>                                                                                                                                             |
| 32 | SAMA: amortization table must specify all payment amounts at signing; declining                        | SAMA Rulebook                    | <a href="https://rulebook.sama.gov.sa/en/key-conditions-calculating-early-repayment-amount">https://rulebook.sama.gov.sa/en/key-conditions-calculating-early-repayment-amount</a>                                                                                                                   |

| #  | What it sources                                                                                  | Publisher        | URL                                                                                                                                                                                       |
|----|--------------------------------------------------------------------------------------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | balance<br>mandatory                                                                             |                  |                                                                                                                                                                                           |
| 33 | SAMA: early<br>prepayment<br>allowed in<br>multiples of full<br>installments;<br>Term Cost fixed | SAMA<br>Rulebook | <a href="https://rulebook.sama.gov.sa/en/article-11-early-payments">https://rulebook.sama.gov.sa/en/article-11-early-payments</a>                                                         |
| 34 | Credit<br>Information Law<br>— prohibits<br>unlicensed<br>credit<br>information<br>services      | SAMA<br>Rulebook | <a href="https://rulebook.sama.gov.sa/en/credit-information-law">https://rulebook.sama.gov.sa/en/credit-information-law</a>                                                               |
| 35 | Riyad Bank<br>Open Banking<br>page (TPP<br>registration<br>mandatory)                            | Riyad Bank       | <a href="https://www.riyadbank.com/open-banking">https://www.riyadbank.com/open-banking</a>                                                                                               |
| 36 | SAMA<br>Responsible<br>Lending<br>Principles<br>(SIMAH<br>mandatory)                             | SAMA<br>Rulebook | <a href="https://rulebook.sama.gov.sa/en/responsible-lending-principles-individual-customers-0">https://rulebook.sama.gov.sa/en/responsible-lending-principles-individual-customers-0</a> |
| 37 | Legal500 Saudi<br>Fintech 2025-<br>2026 (PDPL<br>enforcement,<br>TPP licensing)                  | Legal500         | <a href="https://www.legal500.com/guides/chapter/saudi-arabia-fintech/">https://www.legal500.com/guides/chapter/saudi-arabia-fintech/</a>                                                 |

| #  | What it sources                                                                     | Publisher     | URL                                                                                                                                                                                                                                     |
|----|-------------------------------------------------------------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 38 | SAMA Regulatory Sandbox Framework (always-open, 6-month review, 6-12 month testing) | SAMA Rulebook | <a href="https://rulebook.sama.gov.sa/en/entiresection/1368">https://rulebook.sama.gov.sa/en/entiresection/1368</a>                                                                                                                     |
| 39 | SAMA sandbox: 6-month application review, testing 6-12 months                       | Pontinova Law | <a href="https://www.pontinova.law/blog-posts/saudi-arabia-vision-2030-regulatory-sandbox">https://www.pontinova.law/blog-posts/saudi-arabia-vision-2030-regulatory-sandbox</a>                                                         |
| 40 | SAMA Open Banking Lab (sandbox for API testing and certification)                   | SAMA          | <a href="https://openbanking.sa/index-en.html">https://openbanking.sa/index-en.html</a>                                                                                                                                                 |
| 41 | 19 fintechs in SAMA sandbox by mid-2025 refining credit decisioning                 | Brankas       | <a href="https://blog.brankas.com/inside-saudi-arabia-open-finance-revolution">https://blog.brankas.com/inside-saudi-arabia-open-finance-revolution</a>                                                                                 |
| 42 | SAMA licensed Tamawal for finance aggregation via sandbox (2024)                    | Ozone API     | <a href="https://ozoneapi.com/the-open-finance-tracker/library/ksa-open-banking-standard/">https://ozoneapi.com/the-open-finance-tracker/library/ksa-open-banking-standard/</a>                                                         |
| 43 | SAMA Rule 300-1-1-4 — freelancer accounts MANDATORY within 1 working day            | SAMA Rulebook | <a href="https://rulebook.sama.gov.sa/en/300114-freelance-job-permit-holder">https://rulebook.sama.gov.sa/en/300114-freelance-job-permit-holder</a>                                                                                     |
| 44 | SAMA Circular — all banks must accept                                               | SAMA Rulebook | <a href="https://rulebook.sama.gov.sa/en/adopting-self-employment-document-and-enabling-its-holder-open-bank-accounts">https://rulebook.sama.gov.sa/en/adopting-self-employment-document-and-enabling-its-holder-open-bank-accounts</a> |



| #  | What it sources                                                                 | Publisher                      | URL                                                                                                                                                                                                                                                                                 |
|----|---------------------------------------------------------------------------------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | freelancer<br>accounts within<br>1 working day                                  |                                |                                                                                                                                                                                                                                                                                     |
| 45 | PDPL enforcement — 48 decisions in 2025, fines up to SAR 5M, criminal penalties | A&O Shearman (Apr 2026)        | <a href="https://www.aoshearman.com/en/insights/enforcement-of-the-saudi-personal-data-protection-law">https://www.aoshearman.com/en/insights/enforcement-of-the-saudi-personal-data-protection-law</a>                                                                             |
| 46 | SDAIA — 48 enforcement decisions confirmed, active enforcement phase            | IAPP (Feb 2026)                | <a href="https://iapp.org/news/a/saudi-arabia-s-data-protection-authority-steps-up-enforcement">https://iapp.org/news/a/saudi-arabia-s-data-protection-authority-steps-up-enforcement</a>                                                                                           |
| 47 | PDPL cross-border transfer violation: 1 year imprisonment + SAR 1M fine         | Global Privacy Blog            | <a href="https://www.globalprivacyblog.com/2024/03/six-months-until-enforcement-key-compliance-steps-for-saudi-arabias-data-protection-law/">https://www.globalprivacyblog.com/2024/03/six-months-until-enforcement-key-compliance-steps-for-saudi-arabias-data-protection-law/</a> |
| 48 | PDPL penalties: up to SAR 5M fine, doubling for repeat offences                 | ICLG Data Protection 2025-2026 | <a href="https://iclg.com/practice-areas/data-protection-laws-and-regulations/saudi-arabia">https://iclg.com/practice-areas/data-protection-laws-and-regulations/saudi-arabia</a>                                                                                                   |

**URL verification note:** Verified May 17, 2026. Saudi government and media sites return HTTP 403 to automated curl requests due to standard bot-protection — this does NOT indicate broken links. All URLs confirmed accessible via browser and search result retrieval.

*Document v6 — final regulatory update. SAMA Rule 300-1-1-4 upgraded from optional to mandatory (one working day). PDPL enforcement confirmed active: 48 decisions in 2025. Data residency elevated to criminal liability (1 year imprisonment). 33.33% income formula marked as critical pilot blocker. Section 10 (Critical Pre-Pilot Requirements) added. All (---) page break dividers removed except between major sections. 48 sources. Companion documents: Mihan\_Regulatory\_Verification.md · Mihan\_Final\_Regulatory\_Clearance.md*